

```

DROP TABLE Artist CASCADE CONSTRAINTS;
DROP TABLE Venue CASCADE CONSTRAINTS;
DROP TABLE Fan CASCADE CONSTRAINTS;
DROP TABLE MERCHITEM CASCADE CONSTRAINTS;
DROP TABLE Tour CASCADE CONSTRAINTS;
DROP TABLE CONCERT CASCADE CONSTRAINTS;
DROP TABLE MERCHTRANSACTION CASCADE CONSTRAINTS;
DROP TABLE TICKETTRANSACTION CASCADE CONSTRAINTS;
DROP TABLE MERCHLINEITEM CASCADE CONSTRAINTS;

```

-- STEP 1: Create Master (Parent) Tables

```

CREATE TABLE Artist (
    ArtistID   VARCHAR2(50) CONSTRAINT Artist_PK PRIMARY KEY,
    ArtistName VARCHAR2(50) CONSTRAINT ArtistNameNotNull NOT NULL,
    ArtistGenre VARCHAR2(50) CONSTRAINT ArtistGenreNotNull NOT NULL
);

```

```

CREATE TABLE Venue (
    VenueID      VARCHAR2(50) CONSTRAINT VENUE_PK PRIMARY KEY,
    SeatingCapacity INTEGER    CONSTRAINT CapacityNotNull NOT NULL,
    Location     VARCHAR2(150) CONSTRAINT LocationNotNull NOT NULL,
    Name         VARCHAR2(50)  CONSTRAINT VenueNameNotNull NOT NULL
);

```

```

CREATE TABLE Fan (
    FanID       VARCHAR2(100) CONSTRAINT Fan_PK PRIMARY KEY,
    PhoneNumber NUMBER(10)   CONSTRAINT FanPhoneUnique UNIQUE NOT NULL,
    FanEmail    VARCHAR2(100) CONSTRAINT FanEmailUnique UNIQUE NOT NULL,
    FanCC       NUMBER(16)   CONSTRAINT FanCCNotNull NOT NULL
);

```

```

CREATE TABLE MERCHITEM (
    Ensure merch isn't free and inventory isn't negative.
    CREATE TABLE MERCHITEM (
        SKU          VARCHAR2(50) CONSTRAINT MERCHITEM_PK PRIMARY KEY,
        ItemType     VARCHAR2(100) CONSTRAINT ItemTypeNN NOT NULL,
        MerchPrice    NUMBER(10,2) CONSTRAINT PricePositive CHECK (MerchPrice > 0), --
        Price must be > 0
        QuantityAvailable INTEGER    CONSTRAINT InventoryNonNeg CHECK (QuantityAvailable
        >= 0) -- Inventory can be 0 but not -1
    );

```

-- STEP 2: Create Child Tables (with Foreign Keys)

```
CREATE TABLE Tour (  
    TourID      VARCHAR2(50) CONSTRAINT Tour_PK PRIMARY KEY,  
    NumberOfShows INTEGER    CONSTRAINT ShowsNotNull NOT NULL,  
    ArtistID     VARCHAR2(50) CONSTRAINT TourArtistNotNull NOT NULL,  
    CONSTRAINT Tour_Artist_FK FOREIGN KEY (ArtistID) REFERENCES Artist(ArtistID)  
);
```

```
CREATE TABLE CONCERT (  
    ConcertID   VARCHAR2(50) CONSTRAINT CONCERT_PK PRIMARY KEY,  
    ConcertTime TIMESTAMP    CONSTRAINT ConcertTimeNN NOT NULL,  
    ConcertDate DATE          CONSTRAINT ConcertDateNN NOT NULL,  
    Tour_TourID VARCHAR2(50) CONSTRAINT ConcertTourNN NOT NULL,  
    VENUE_VenueID VARCHAR2(50) CONSTRAINT ConcertVenueNN NOT NULL,  
    CONSTRAINT CONCERT_Tour_FK FOREIGN KEY (Tour_TourID) REFERENCES  
Tour(TourID),  
    CONSTRAINT CONCERT_Venue_FK FOREIGN KEY (VENUE_VenueID) REFERENCES  
Venue(VenueID)  
);
```

```
CREATE TABLE MERCHTRANSACTION (  
    OrderNumber VARCHAR2(100) CONSTRAINT MERCHTRANSACTION_PK PRIMARY KEY,  
    OrderDate   DATE          CONSTRAINT OrderDateNotNull NOT NULL,  
    OrderTotal  NUMBER(10,2)  CONSTRAINT OrderTotalNotNull NOT NULL,  
    Fan_FanID   VARCHAR2(100) CONSTRAINT TransactionFanNotNull NOT NULL,  
    CONSTRAINT MERCHTRANSACTION_Fan_FK FOREIGN KEY (Fan_FanID)  
REFERENCES Fan(FanID)  
);
```

```
CREATE TABLE TICKETTRANSACTION (  
    TicketID     VARCHAR2(50) CONSTRAINT Ticket_PK PRIMARY KEY,  
    Section      INTEGER    CONSTRAINT TicketSection_NN NOT NULL,  
    VenueRow     VARCHAR2(3)  CONSTRAINT TicketRow_NN NOT NULL, -- Renamed from  
'Row'  
    Seat         INTEGER     CONSTRAINT TicketSeat_NN NOT NULL,  
    CONCERT_ConcertID VARCHAR2(50) CONSTRAINT TicketConcert_NN NOT NULL,  
    Fan_FanID     VARCHAR2(100) CONSTRAINT TicketFan_NN NOT NULL,  
    CONSTRAINT Ticket_Concert_FK FOREIGN KEY (CONCERT_ConcertID)
```

```

REFERENCES CONCERT(ConcertID),
CONSTRAINT Ticket_Fan_FK FOREIGN KEY (Fan_FanID)
REFERENCES Fan(FanID)
);

-- STEP 3: Create Associative Table (Composite Primary Key)
CREATE TABLE MERCHLINEITEM (
  MERCHTRANSACTION_OrderID VARCHAR2(100), -- Defined here
  MERCHITEM_ItemID          VARCHAR2(50),
  SalePrice                 NUMBER(10,2) CONSTRAINT SalePriceNotNull NOT NULL,
  AmtBought                 INTEGER    CONSTRAINT AmtBoughtNotNull NOT NULL,
  -- Match the names below to 'MERCHTRANSACTION_OrderID'
  CONSTRAINT MERCHLINEITEM_PK PRIMARY KEY (MERCHTRANSACTION_OrderID,
  MERCHITEM_ItemID),
  CONSTRAINT MERCHLINEITEM_MERCHTRANSACTION_FK FOREIGN KEY
(MERCHTRANSACTION_OrderID)
  REFERENCES MERCHTRANSACTION(OrderNumber),
  CONSTRAINT MERCHLINEITEM_MERCHITEM_FK FOREIGN KEY (MERCHITEM_ItemID)
  REFERENCES MERCHITEM(SKU)
);

```

```

-- 1. ARTIST: singular-sounding fake names with "A0#" IDs
INSERT INTO Artist VALUES ('A01', 'Rowan Thorne', 'Indie Folk');
INSERT INTO Artist VALUES ('A02', 'Elias Vance', 'Alternative Rock');
INSERT INTO Artist VALUES ('A03', 'Silas Moon', 'Synth-Pop');
INSERT INTO Artist VALUES ('A04', 'Juniper Vale', 'Acoustic');
INSERT INTO Artist VALUES ('A05', 'Ezra Sterling', 'Neo-Soul');
INSERT INTO Artist VALUES ('A06', 'Maya Rhodes', 'Americana');
INSERT INTO Artist VALUES ('A07', 'Julian Grey', 'Rhythm and Blues');
INSERT INTO Artist VALUES ('A08', 'Clara West', 'Country');
INSERT INTO Artist VALUES ('A09', 'Leo Blackwood', 'Blues');
INSERT INTO Artist VALUES ('A10', 'Ivy Monroe', 'Jazz');

```

```

-- 2. VENUE
INSERT INTO Venue VALUES ('V01', 500, 'Nashville, TN', 'The Soundbox');
INSERT INTO Venue VALUES ('V02', 1200, 'Austin, TX', 'Neon Garden');
INSERT INTO Venue VALUES ('V03', 300, 'Seattle, WA', 'Emerald Stage');
INSERT INTO Venue VALUES ('V04', 2500, 'Chicago, IL', 'Windy City Hall');
INSERT INTO Venue VALUES ('V05', 800, 'Denver, CO', 'Peak Pavilion');
INSERT INTO Venue VALUES ('V06', 1500, 'Atlanta, GA', 'Peach Tree Arena');

```

```
INSERT INTO Venue VALUES ('V07', 400, 'Portland, OR', 'Rose City Club');
INSERT INTO Venue VALUES ('V08', 5000, 'New York, NY', 'Liberty Square');
INSERT INTO Venue VALUES ('V09', 1000, 'Boston, MA', 'Old Harbor Theater');
INSERT INTO Venue VALUES ('V10', 2000, 'Miami, FL', 'Tropical Sands');
```

-- 3. FAN: Creative area codes and unique phone numbers

```
INSERT INTO Fan VALUES ('F01', 3055550199, 'j.doe@gmail.com', 1234567812345678);
INSERT INTO Fan VALUES ('F02', 2125550123, 'smith.a@yahoo.com', 8765432187654321);
INSERT INTO Fan VALUES ('F03', 4155550177, 'mike.j@outlook.com', 1111222233334444);
INSERT INTO Fan VALUES ('F04', 6175550144, 'sarah.v@gmail.com', 5555666677778888);
INSERT INTO Fan VALUES ('F05', 3125550188, 'c.wilson@me.com', 9999000011112222);
INSERT INTO Fan VALUES ('F06', 2065550111, 'amy_lee@edu.com', 3333444455556666);
INSERT INTO Fan VALUES ('F07', 7025550155, 'b_robinson@gmail.com',
7777888899990000);
INSERT INTO Fan VALUES ('F08', 5035550166, 't_miller@work.com', 1212232334344545);
INSERT INTO Fan VALUES ('F09', 6155550133, 'lisa_g@mail.com', 5656676778788989);
INSERT INTO Fan VALUES ('F10', 4045550122, 'dave_k@gmail.com', 9090010112122323);
```

-- 4. MERCHITEM: 6-digit grocery-style SKUs

```
INSERT INTO MERCHITEM VALUES ('481023', 'T-Shirt', 'Medium', 25.00, 100);
INSERT INTO MERCHITEM VALUES ('481024', 'Hoodie', 'Large', 55.00, 50);
INSERT INTO MERCHITEM VALUES ('481025', 'Poster', 'OS', 15.00, 200);
INSERT INTO MERCHITEM VALUES ('481026', 'Vinyl', 'Standard', 35.00, 30);
INSERT INTO MERCHITEM VALUES ('481027', 'Cap', 'OS', 20.00, 75);
INSERT INTO MERCHITEM VALUES ('481028', 'Tote Bag', 'OS', 12.00, 150);
INSERT INTO MERCHITEM VALUES ('481029', 'Sticker Pack', 'OS', 5.00, 500);
INSERT INTO MERCHITEM VALUES ('481030', 'Long Sleeve', 'XL', 40.00, 40);
INSERT INTO MERCHITEM VALUES ('481031', 'Beanie', 'OS', 18.00, 60);
INSERT INTO MERCHITEM VALUES ('481032', 'Keychain', 'OS', 8.00, 300);
```

-- 5. TOUR

```
INSERT INTO Tour VALUES ('T01', 12, 'A01');
INSERT INTO Tour VALUES ('T02', 8, 'A02');
INSERT INTO Tour VALUES ('T03', 15, 'A03');
INSERT INTO Tour VALUES ('T04', 10, 'A04');
INSERT INTO Tour VALUES ('T05', 20, 'A05');
INSERT INTO Tour VALUES ('T06', 5, 'A06');
INSERT INTO Tour VALUES ('T07', 25, 'A07');
INSERT INTO Tour VALUES ('T08', 14, 'A08');
INSERT INTO Tour VALUES ('T09', 18, 'A09');
INSERT INTO Tour VALUES ('T10', 30, 'A10');
```

-- 6. CONCERT

```
INSERT INTO CONCERT VALUES ('C01', to_timestamp('20:00:00', 'HH24:MI:SS'),
to_date('06/15/2026', 'MM/DD/YYYY'), 'T01', 'V01');
INSERT INTO CONCERT VALUES ('C02', to_timestamp('19:30:00', 'HH24:MI:SS'),
to_date('06/18/2026', 'MM/DD/YYYY'), 'T02', 'V02');
INSERT INTO CONCERT VALUES ('C03', to_timestamp('21:00:00', 'HH24:MI:SS'),
to_date('06/22/2026', 'MM/DD/YYYY'), 'T03', 'V03');
INSERT INTO CONCERT VALUES ('C04', to_timestamp('20:00:00', 'HH24:MI:SS'),
to_date('06/25/2026', 'MM/DD/YYYY'), 'T04', 'V04');
INSERT INTO CONCERT VALUES ('C05', to_timestamp('19:00:00', 'HH24:MI:SS'),
to_date('06/28/2026', 'MM/DD/YYYY'), 'T05', 'V05');
INSERT INTO CONCERT VALUES ('C06', to_timestamp('20:30:00', 'HH24:MI:SS'),
to_date('07/02/2026', 'MM/DD/YYYY'), 'T06', 'V06');
INSERT INTO CONCERT VALUES ('C07', to_timestamp('21:30:00', 'HH24:MI:SS'),
to_date('07/05/2026', 'MM/DD/YYYY'), 'T07', 'V07');
INSERT INTO CONCERT VALUES ('C08', to_timestamp('19:00:00', 'HH24:MI:SS'),
to_date('07/10/2026', 'MM/DD/YYYY'), 'T08', 'V08');
INSERT INTO CONCERT VALUES ('C09', to_timestamp('20:00:00', 'HH24:MI:SS'),
to_date('07/14/2026', 'MM/DD/YYYY'), 'T09', 'V09');
INSERT INTO CONCERT VALUES ('C10', to_timestamp('22:00:00', 'HH24:MI:SS'),
to_date('07/20/2026', 'MM/DD/YYYY'), 'T10', 'V10');
```

-- 7. MERCHTRANSACTION: Spread-out order dates

```
INSERT INTO MERCHTRANSACTION VALUES ('ORD01', to_date('05/01/2026',
'MM/DD/YYYY'), 25.00, 'F01');
INSERT INTO MERCHTRANSACTION VALUES ('ORD02', to_date('05/05/2026',
'MM/DD/YYYY'), 110.00, 'F02');
INSERT INTO MERCHTRANSACTION VALUES ('ORD03', to_date('05/12/2026',
'MM/DD/YYYY'), 15.00, 'F03');
INSERT INTO MERCHTRANSACTION VALUES ('ORD04', to_date('05/20/2026',
'MM/DD/YYYY'), 35.00, 'F04');
INSERT INTO MERCHTRANSACTION VALUES ('ORD05', to_date('06/02/2026',
'MM/DD/YYYY'), 40.00, 'F05');
INSERT INTO MERCHTRANSACTION VALUES ('ORD06', to_date('06/15/2026',
'MM/DD/YYYY'), 12.00, 'F06');
INSERT INTO MERCHTRANSACTION VALUES ('ORD07', to_date('06/28/2026',
'MM/DD/YYYY'), 10.00, 'F07');
INSERT INTO MERCHTRANSACTION VALUES ('ORD08', to_date('07/04/2026',
'MM/DD/YYYY'), 80.00, 'F08');
INSERT INTO MERCHTRANSACTION VALUES ('ORD09', to_date('07/18/2026',
'MM/DD/YYYY'), 18.00, 'F09');
```

```
INSERT INTO MERCHTRANSACTION VALUES ('ORD10', to_date('08/05/2026',  
'MM/DD/YYYY'), 16.00, 'F10');
```

```
-- 8. TICKETTRANSACTION
```

```
INSERT INTO TICKETTRANSACTION VALUES ('TIX01', 101, 'A', 12, 'C01', 'F01');  
INSERT INTO TICKETTRANSACTION VALUES ('TIX02', 102, 'B', 5, 'C02', 'F02');  
INSERT INTO TICKETTRANSACTION VALUES ('TIX03', 201, 'F', 22, 'C03', 'F03');  
INSERT INTO TICKETTRANSACTION VALUES ('TIX04', 105, 'D', 1, 'C04', 'F04');  
INSERT INTO TICKETTRANSACTION VALUES ('TIX05', 300, 'G', 30, 'C05', 'F05');  
INSERT INTO TICKETTRANSACTION VALUES ('TIX06', 110, 'C', 15, 'C06', 'F06');  
INSERT INTO TICKETTRANSACTION VALUES ('TIX07', 112, 'E', 8, 'C07', 'F07');  
INSERT INTO TICKETTRANSACTION VALUES ('TIX08', 205, 'B', 19, 'C08', 'F08');  
INSERT INTO TICKETTRANSACTION VALUES ('TIX09', 101, 'A', 3, 'C09', 'F09');  
INSERT INTO TICKETTRANSACTION VALUES ('TIX10', 400, 'Z', 10, 'C10', 'F10');
```

```
-- 9. MERCHLINEITEM: Linking to spread dates and grocery-style SKUs
```

```
INSERT INTO MERCHLINEITEM VALUES ('ORD01', '481023', 25.00, 1);  
INSERT INTO MERCHLINEITEM VALUES ('ORD02', '481024', 55.00, 2);  
INSERT INTO MERCHLINEITEM VALUES ('ORD03', '481025', 15.00, 1);  
INSERT INTO MERCHLINEITEM VALUES ('ORD04', '481026', 35.00, 1);  
INSERT INTO MERCHLINEITEM VALUES ('ORD05', '481027', 20.00, 2);  
INSERT INTO MERCHLINEITEM VALUES ('ORD06', '481028', 12.00, 1);  
INSERT INTO MERCHLINEITEM VALUES ('ORD07', '481029', 5.00, 2);  
INSERT INTO MERCHLINEITEM VALUES ('ORD08', '481030', 40.00, 2);  
INSERT INTO MERCHLINEITEM VALUES ('ORD09', '481031', 18.00, 1);  
INSERT INTO MERCHLINEITEM VALUES ('ORD10', '481032', 8.00, 2);
```

```
--complex queries
```

```
--subquery
```

```
SELECT Fan_FanID, OrderNumber, OrderTotal  
FROM MERCHTRANSACTION  
WHERE OrderTotal > (SELECT AVG(OrderTotal) FROM MERCHTRANSACTION);
```

```
--inner join
```

```
SELECT A.ArtistName, T.TourID, T.NumberOfShows  
FROM Artist A  
INNER JOIN Tour T ON A.ArtistID = T.ArtistID;
```

```
--outer join
```

```
SELECT V.Name, C.ConcertID, C.ConcertDate  
FROM Venue V
```

```
LEFT OUTER JOIN CONCERT C ON V.VenueID = C.VENUE_VenueID;
```

```
--group by no having  
SELECT ArtistGenre, COUNT(*) AS TotalArtists  
FROM Artist  
GROUP BY ArtistGenre;
```

```
--group by with having  
SELECT ItemType, SUM(SalePrice * AmtBought) AS TotalRevenue  
FROM MERCHITEM M  
JOIN MERCHLINEITEM MLI ON M.SKU = MLI.MERCHITEM_ItemID  
GROUP BY ItemType  
HAVING SUM(SalePrice * AmtBought) > 100.00;
```

```
--relevant index  
CREATE INDEX MerchDateIndex ON MERCHTRANSACTION (OrderDate);
```

```
--relevantView  
CREATE VIEW ConcertMasterList AS  
SELECT A.ArtistName, V.Name AS VenueName, C.ConcertDate, C.ConcertTime  
FROM Artist A  
JOIN Tour T ON A.ArtistID = T.ArtistID  
JOIN CONCERT C ON T.TourID = C.Tour_TourID  
JOIN Venue V ON C.VENUE_VenueID = V.VenueID;
```

```
--Relevant trigger  
CREATE OR REPLACE TRIGGER UpdateMerchInventory  
AFTER INSERT ON MERCHLINEITEM  
FOR EACH ROW  
BEGIN  
    UPDATE MERCHITEM  
    SET QuantityAvailable = QuantityAvailable - :NEW.AmtBought  
    WHERE SKU = :NEW.MERCHITEM_ItemID;  
END;
```